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TransFGVC: transformer-based fine-grained visual classification

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Abstract

Fine-grained visual classification (FGVC) aims to identify subcategories of objects within the same superclass. This task is challenging owing to high intra-class variance and low inter-class variance. The most recent methods focus on locating discriminative areas and then training the classification network to further capture the subtle differences among them. On the one hand, the detection network often obtains an entire part of the object, and positioning errors occur. On the other hand, these methods ignore the correlations between the extracted regions. We propose a novel highly scalable approach, called TransFGVC, that cleverly combines Swin Transformers with long short-term memory (LSTM) networks to address the above problems. The Swin Transformer is used to obtain remarkable visual tokens through self-attention layer stacking, and LSTM is used to model them globally, which not only accurately locates the discriminative region but also further introduces global information that is important for FGVC. The proposed method achieves competitive performance with accuracy rates of 92.7%, 91.4% and 91.5% using the public CUB-200-2011 and NABirds datasets and our Birds-267-2022 dataset, and the Params and FLOPs of our method are 25% and 27% lower, respectively, than the current SotA method HERBS. To effectively promote the development of FGVC, we developed the Birds-267-2022 dataset, which has 267 categories and 12,233 images.

Keywords Computer vision · Fine-grained visual classification · LSTM · Swin Transformer · Birds-267-2022 dataset

1 Introduction

When we examine an image, we can discern whether it is a car or an airplane, as well as whether a dog is a Welsh Corgi or Golden Retriever. This is because of the superior reasoning ability of the human visual system. In context of the examples above, Fig. 1a illustrates generic image classification and Fig. 1b illustrates fine-grained visual classification (FGVC). The goal of FGVC is to distinguish subclasses of objects from the same superclass. Moreover, FGVC has been widely applied in industry and daily life, such as intelligent retail [1], intelligent transportation [2] and automatic biodiversity monitoring [3], producing a positive impact on human society. Many related datasets have been proposed to train

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Fig. 1 Generic image analysis versus fine-grained image analysis



machines to view images in a fine-grained manner. FGVC is more challenging than generic image recognition for the following reasons.

- High intra-class variance: owing to the poses, scales, viewpoints and illuminations, the same class of objects often have significant visual differences (Fig. 2).
- Low inter-class variance: objects from different categories are extremely similar in color and shape; therefore, it is difficult to distinguish them (Fig. 3).
- Limited datasets: classical fine-grained datasets such as CUB-200-2011, FGVC-Aircraft, Stanford Cars and Stanford Dogs are used for benchmarking FGVC. However, according to modern standards, these datasets are relatively small and highly saturated in terms of performance (Table 1). Therefore, more challenging datasets must be proposed.

Therefore, FGVC has piqued considerable attention of researchers [7, 16–26]. Many excellent research results have been obtained in the field of FGVC, with some initial methods [27–29] localizing objects and aligning related parts by incorporating additional bounding box/part annotations. However, collecting additional annotation information is not only laborious, but also requires expertise in specific fields, which makes these methods impractical. Therefore, FGVC based on weak supervision has attracted more attention from researchers, as it only needs image labels for supervision.

For this purpose, two main paradigms have developed. One paradigm is the localization classification subnetwork. These methods based on this [30–32] mainly use Region Proposal Networks (RPNs) to propose the bounding box of the regions of interest containing the distinguishing parts. Then

the selected image areas are readjusted to a predetermined size and forwarded again through the backbone network to obtain rich local features.

In summary, a common strategy is to use RPNs for object localization, use the regions of interest to classify individually and apply a rank loss [33] to maintain balance between the final classification output and the quality of bounding boxes. However, this method may isolate the selected regions and ignore the relationship among selected regions. To obtain correct classification results, the RPN sometimes inevitably proposes large bounding boxes that may contain most parts of the objects. In addition, obtained bounding boxes may contain an exceedingly large background area or may not even localize object parts, leading to confusion. Moreover, compared to the backbone network, an RPN module with different optimizing goals may render the network more difficult to train, and the reuse of the backbone network makes the algorithm more complicated.

The other paradigm is end-to-end feature encoding [34–36], which extracts the fine-grained features by encoding a highly parameterized representation. These methods aggregate pairs of related local descriptors into a global representation of features to model the second-order information of input descriptions, thus producing better distinction and representation. However, the high-dimensional bilinear features cause visual burstiness problems, redundant features, overfitting and a significant computational burden.

Vision transformer (ViT)-based methods have recently become popular, excelling in vision tasks such as image classification, segmentation and detection. Recently, many ViT-based methods have started to emerge in the field of FGVC. ViT demonstrates notable advantages for the aforementioned tasks. On the one hand, we believe that this is a good start. On the other hand, it is still an open issue to make

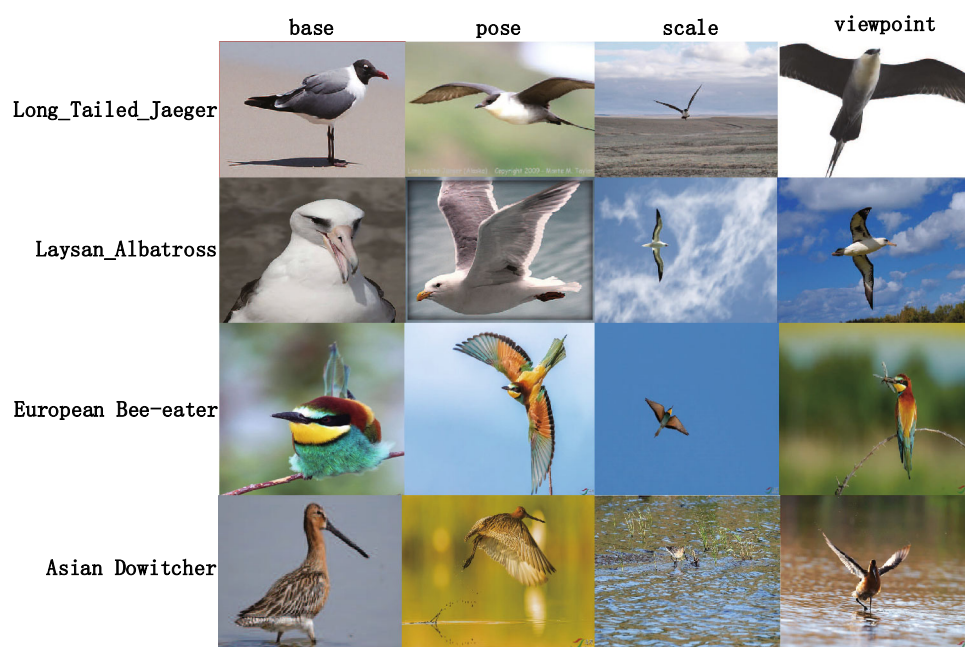
Fig. 2 High intra-class variance**Fig. 3** Low inter-class variance

Table 1 Detailed introduction to public typical datasets in FGVC

Dataset name	Year	Meta-class	#Train/#Test	Categories	The best accuracy to date (%)
Oxford 102 Flower [4]	2008	Flower	2040/6149	102	99.6 (IELT. Xu et al. [5])
CUB-200-2011 [6]	2011	Bird	5994/5794	200	93.1 (HERBS. Chou et al. [7])
Stanford Dogs [8]	2011	Dog	12,000/8580	120	97.1 (WS-CPM. Ge et al. [9])
Oxford-IIIT Pets [10]	2012	Cat, Dog	3680/3669	37	97.3 (CAP. Behera et al. [11])
Stanford Cars [12]	2013	Car	8144/8041	196	97.1 (CMAL-Net. Liu et al. [13])
FGVC-Aircraft [14]	2013	Aircraft	6667/3333	100	94.9 (CAP. Behera et al. [11])
NABirds [3]	2015	Bird	23,929/24,633	555	93.0 (HERBS. Chou et al. [7])
Food-101 [15]	2017	Food	75,750/25,250	101	98.6 (CAP. Behera et al. [11])

full use of the distinctive local features obtained through self-attention for suitable application to FGVC. In addition, these ViT-based jobs impose a significant computational burden and require further optimization.

In this study, we propose a general solution for FGVC based on the Swin Transformer [37] by combining it with a long short-term memory (LSTM) network. In particular, we completely utilize the local self-attention in the Swin Transformer to extract the distinguishable local features of objects and further re-encode and combine the contextual features with LSTM. The experiments obtained excellent results on the public CUB-200-2011, NABirds and the Birds-267-2022 dataset.

In summary, the main contributions of this study are summarized as follows.

- To solve the problem of high intra-class variance and low inter-class variance, we propose a novel Swin Transformer-based FGVC approach. The Swin Transformer is unified with LSTM to enhance the global modeling capabilities of the model through feature enhancement and fusion.
- To solve the problem of a limited dataset, we provide a new fine-grained dataset called Birds-267-2022 for FGVC. We hope that this dataset will promote research progress for deep learning techniques in FGVC.
- Extensive experiments demonstrate the effectiveness of the proposed approach against the state-of-the-art (SotA) FGVC methods on the public CUB-200-2011, NABirds and Birds-267-2022 benchmark datasets.

To better illustrate the effect of the method and experiments proposed in this study, the remainder of this paper is structured as follows. Section 2 provides a review of related work, and Sect. 3 describes our method in detail. Section 4 presents the proposed dataset. Section 5 describes the experimental results and performance analysis, followed by conclusion in Sect. 6.

2 Related work

In this section, we briefly review the representative studies in FGVC and the studies related to our method.

2.1 Methods based on localization

Many methods based on localization classification subnetworks exist, which can be divided into two types: strong supervision [27, 38, 39] and weak supervision [16, 30, 40]. For example, Zhang et al. [27], first generated part- and object-level bounding boxes based on annotations. Subsequently, an R-CNN [41] model was trained to perform object and part detection. Finally, the obtained image patch was used as the input, and a convolutional neural network (CNN) was trained for the classification task. Furthermore, Branson et al. [38] proposed use of groups of detected part key points for computing different levels of warped image patches, before corresponding part-level features were obtained by pose normalization. They also suggested that the convolution features of different layers should be extracted for different levels of the patch. Finally, different levels of features were cascaded as the representations of the entire image. Zheng et al. [40] introduced a multi-attention CNN (MA-CNN)-based part learning method that enables part generation and feature learning to complement mutually. Ding et al. [30] proposed an attention pyramid network by developing a dual-pathway hierarchy to generate the pyramidal regions of interest. After refining the regions of interest, they generated predictions.

These methods provide valuable information for current research and significantly promote the development of FGVC. However, the aforementioned methods either use extra annotations in the training process or require the training of an additional localization network to identify discriminative parts, which significantly increases the complexity of the network. Therefore, they are not conducive to the application of FGVC to the real world, and the proposed approach is distinguished from the aforementioned methods.

2.2 Methods based on feature encoding

As the computing power of devices increases, feature learning-based methods are becoming more popular. Cui et al. [42] suggested a kernel pooling method that can be combined with a CNN to capture arbitrarily ordered and non-linear features via compact feature mapping. Cai et al. [43] proposed a polynomial kernel-based predictor to handle the problem of part interaction and used polynomial descriptors to integrate the features of different layers through kernel fusion. Subsequently, Engin et al. [44] proposed jointly learning the deep local descriptors and kernel matrix-based covariance representation in an end-to-end trainable manner. Recently, to better help the network to learn more diverse and discriminative feature representations, Song et al. [16] introduced a feature boosting and suppression module (FBSM) and a feature diversification module (FDM) based on CNN. FBSM was used to enhance the most discriminative part-specific region in the feature map and suppress it in the later stage, thus directly forcing the subsequent stage to extract other potential parts. FDM was applied to diversify each part-specific representation by aggregating semantically complementary information from other object parts.

Admittedly, these methods have achieved significant advances in FGVC. However, it is not easy for convolution-based structures to capture global information without special design. In contrast with the aforementioned studies, we use a transformer to experiment with capturing the long-term dependencies of features.

2.3 Transformer-based methods

Previous pioneering studies based on transformers in FGVC are extremely informative. He et al. [21] introduced a partial selection module (PSM) to guide the network to accurately obtain discriminative image regions by integrating all of the original attention weights before the last transformer layer into a new attention map. In contrast with [21], Wang et al. [20] proposed a mutual attention weight selection module (MAWSM) to compensate different levels of feature by selectively aggregating important patches from each transformer layer. The integrated patches and the class token were then sent to the last transformer layer for classification. Recently, Chou et al. [17] introduced a new plug-in module, which can be integrated into many existing backbone networks. This module integrated background segmentation and feature fusion and can accurately provide discriminative areas.

The aforementioned methods are similar to the proposed method. These methods have enabled relevant explorations of FGVC using self-attention, with positive results. In contrast with the aforementioned methods, we propose a method combining the Swin Transformer and LSTM, which can

locate distinguishing regions, further enhance and combine them, as well as obtain competitive results.

This study is inspired by Behera et al. [11], wherein context-aware attention was used to capture contextual information and LSTM was applied to learn the subtle changes in the integral regions. Our novel paradigm is a step-forward unified Swin Transformer with LSTM for FGVC. It is advantageous over some existing methods as it uses a single network, the local self-attention mechanism, to extract discriminative local regions and LSTM to further re-encode and combine context information.

3 Proposed method

In this section, we introduce our TransFGVC approach in detail. Details about the Swin Transformer and LSTM are found in appendix.

The overall framework of our model is shown in Fig. 4. It accepts an input image and provides an output as the predicted class label. The overall solution is described as follows. First, we provide M pictures $D = \{D_m | m = 1, \dots, M\}$ as input to the model F with the corresponding label $L = \{L_m | m = 1, \dots, M\}$. Specifically, the model F is divided into three parts:

1. A basic Swin Transformer network F_t with parameter ω_t ;
2. A LSTM model F_l with learnable parameters ω_l ;
3. The final linear classifier F_c with parameter ω_c .

Then, we then aim to train the model $F: \hat{L}_m = F(D_m)$ that minimizes the cross-entropy loss function by updating the network parameters $\omega = \{\omega_t, \omega_l, \omega_c\}$ such that the predicted label $\hat{L}_m$ matches the true label L_m .

3.1 Transformer-based fine-grained visual classification (TransFGVC) architecture

Extracting the distinguishing regions and global modeling are extremely important in FGVC. Additionally, we believe that they can be mutually reinforced. Considering the challenging CUB-200-2011 dataset as an example, more than one distinguishing feature can exist; therefore, the final classification results should also consider the relationship between them. As shown in Fig. 4a, considering the connection between the eyes and periphery simultaneously will achieve better results, which can also alleviate the overfitting phenomenon of the model to a certain extent.

To determine the distinguishing parts of the object, previous methods such as AP-CNN utilize a region proposal generator to select region proposals with high response values on their responding attention maps, where the accuracy of CUB-200-2011 reached 88.4%. To further enhance the

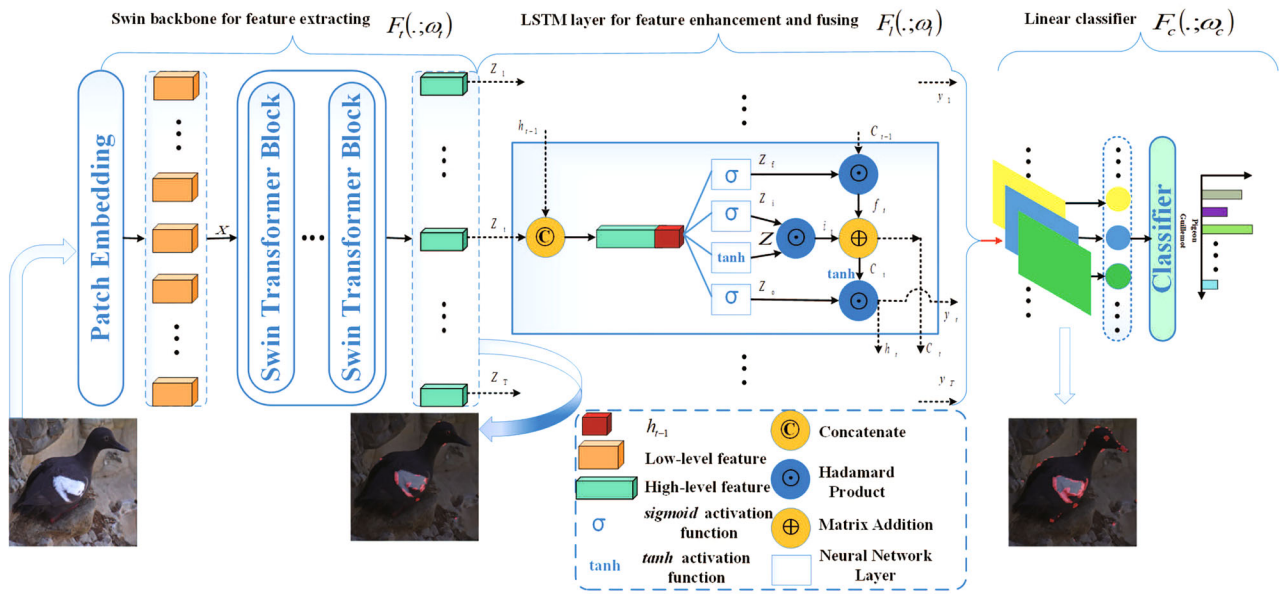


Fig. 4 General framework of the proposed TransFGVC. First, images are split into small patches and then projected into the patch embeddings by linear embedding. The patch embeddings are fed into the Swin Transformer encoder to learn discriminative patch features along with

the relative position embedding. Following the last Swin Transformer layer, an LSTM module is applied to the global modeling of the output distinctive patch via feature enhancement and fusion (this is best viewed in color)

global modeling ability of the model based on the mining of distinctive areas, the FBSM first divides images into simple striped parts to further enhance and suppress different striped parts using a 1×1 convolution, and obtains distinctive regions at different stages. Finally, an FDM module is used to fuse multiple significant regions to improve the global modeling ability of the model, with an accuracy of 89.8%. Experiments have shown that significant progress has been achieved in global modeling. In contrast with the aforementioned methods, we divide the image into fine-grained patches instead of simple striped parts to enable it to be more suitable for FGVC and then mine the fine-grained features of the patch level through local self-attention. In addition, many recent methods based on ViT, such as TransFG, demonstrate that ViT achieves good results in the field of FGVC. This is mainly because ViT-based methods significantly improve the model's ability to capture remote dependencies, achieving an accuracy of 91.7%. In contrast with these studies, our method is based on the Swin Transformer and LSTM to further enhance the global expression ability of the model.

$$\begin{aligned}
 \hat{\mathbf{z}}_l &= \mathbf{W} - \text{MSA}(\text{LN}(\mathbf{z}_{l-1})) + \mathbf{z}_{l-1} \\
 \mathbf{z}_l &= \text{MLP}(\text{LN}(\hat{\mathbf{z}}_l)) + \hat{\mathbf{z}}_l \\
 \hat{\mathbf{z}}_{l+1} &= \text{SW} - \text{MSA}(\text{LN}(\mathbf{z}_l)) + \mathbf{z}_l \\
 \mathbf{z}_{l+1} &= \text{MLP}(\text{LN}(\hat{\mathbf{z}}_{l+1})) + \hat{\mathbf{z}}_{l+1}
 \end{aligned} \quad (1)$$

In this study, the Swin Transformer combined with LSTM functions as follows. First, the Swin Transformer outputs

the sequence tokens $\mathbf{Z} = \{\mathbf{Z}_t \mid t \in [1, T]\}$, as expressed by equation (1). Then, they are sent to the LSTM unit for the fusion and enhancement of context features. We consider the LSTM unit at time t as an example with three inputs. The token input at time t is recorded as $\mathbf{Z}_t \in \mathbb{R}^{s \times m}$, and the cell state $\mathbf{C}_{t-1} \in \mathbb{R}^{s \times n_1}$ and hidden state $\mathbf{h}_{t-1} \in \mathbb{R}^{s \times n_1}$ are output at time $(t-1)$. At the initialization stage, first, $\mathbf{Z}_t$ and $\mathbf{h}_{t-1}$ are combined, and then, four copies of it are recorded as $\mathbf{z}_{t1}, \mathbf{z}_{t2}, \mathbf{z}_{t3}, \mathbf{z}_{t4} \in \mathbb{R}^{s \times (m+n_1)}$. Next, $\mathbf{z}_{t1}, \mathbf{z}_{t2}$, and $\mathbf{z}_{t4}$ pass through the sigmoid layer to obtain gating signals $\mathbf{z}_f, \mathbf{z}_i, \mathbf{z}_o \in \mathbb{R}^{s \times n_1}$ with values reduced to $[0, 1]$. The term $\mathbf{z}_{t3}$ is processed by the tanh layer to obtain the input information $\mathbf{z} \in \mathbb{R}^{s \times n_1}$ with values reduced to $[-1, 1]$, where s denotes batch size, m denotes the dimension of the input matrix $\mathbf{Z}_t$, and n_1 denotes the dimension of the hidden layer. This can be expressed as follows:

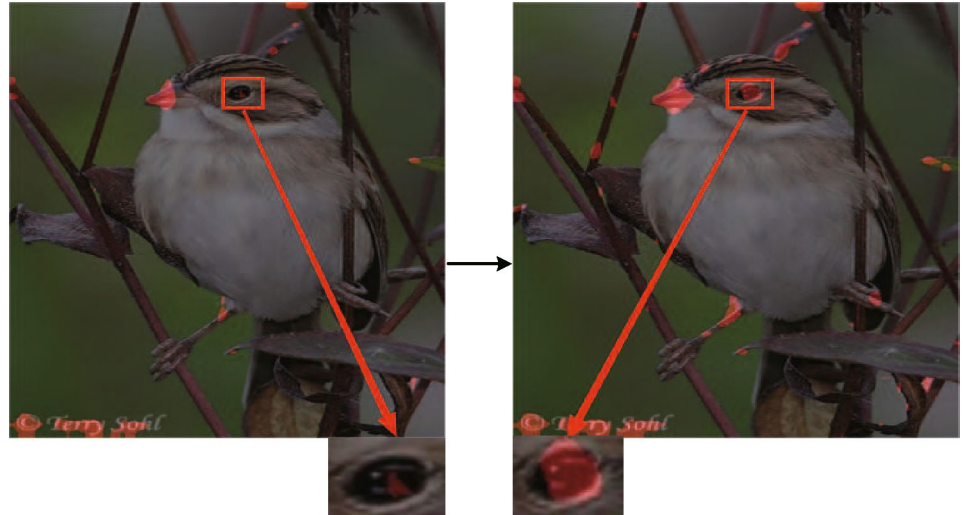
$$\mathbf{z}_{t1}, \mathbf{z}_{t2}, \mathbf{z}_{t3}, \mathbf{z}_{t4} = [\mathbf{Z}_t, \mathbf{C}_{t-1}] \quad (2)$$

$$\mathbf{z}_f, \mathbf{z}_i, \mathbf{z}_o = \sigma(\mathbf{W}_f \cdot \mathbf{z}_{t1} + \mathbf{b}_f), \quad \sigma(\mathbf{W}_i \cdot \mathbf{z}_{t2} + \mathbf{b}_i), \quad (3)$$

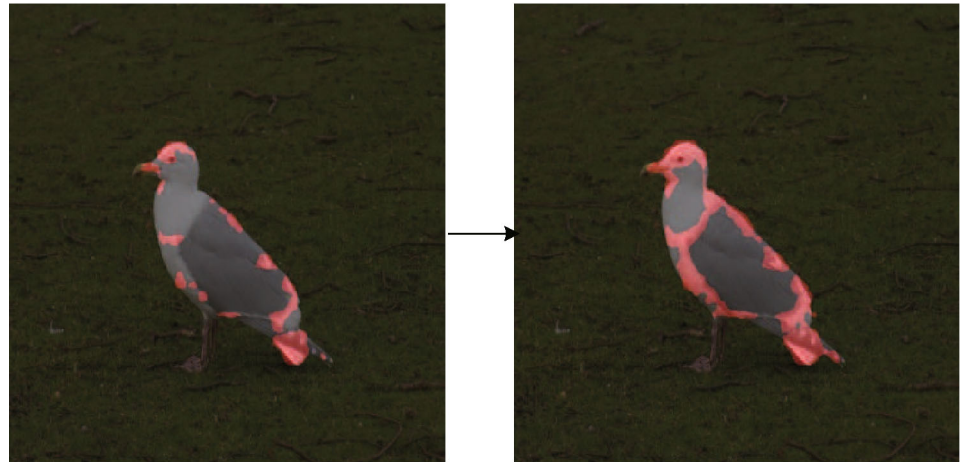
$$\mathbf{z} = \tanh(\mathbf{W}_c \cdot \mathbf{z}_{t3} + \mathbf{b}_c) \quad (4)$$

where $\mathbf{W}_f, \mathbf{W}_i, \mathbf{W}_c, \mathbf{W}_o \in \mathbb{R}^{n_1 \times (m+n_1)}$ and $\mathbf{b}_f, \mathbf{b}_i, \mathbf{b}_c, \mathbf{b}_o \in \mathbb{R}^{n_1}$ are the weight matrixes and the bias of the corresponding positions, respectively. In addition, σ and $\tanh$ represent the sigmoid and hyperbolic tangent activation functions, respec-

Fig. 5 LSTM has the ability to model the target globally. **a** and **b** both have two pictures; the left is the visual result of the Swin Transformer, whereas the right is the visual result of the Swin Transformer with LSTM



(a) LSTM can fuse the eyes and surrounding areas of birds in the Clay-Colored_Sparrow category.



(b) LSTM can enhance the saliency regions of birds in the Glaucous-Winged_Gull category.

tively. After the initialization stage, we summarize the formal process into four steps. First, the gating signal z_f is multiplied by the hidden state C_{t-1} of the element to forget the redundant information in the cell state of the tokens at the previous moment and obtain $f_t \in R^{s \times n_1}$.

$$f_t = z_f \odot C_{t-1} \quad (5)$$

Then, the input information z and the gating signal z_i are multiplied element by element to obtain $i_t \in R^{s \times n_1}$, and the new information passed by the token at this instant is selectively retained.

$$i_t = z \odot z_i \quad (6)$$

Second, the information obtained by the input and forgetting gates is added element by element to obtain the updated cell state $C_t \in R^{s \times n_1}$, which is then sent to the next unit. Notably,

this operation not only focuses on the information of previous nodes but also interacts with the updated information of this node, thus further improving the global modeling ability of the model. We believe that this is an important reason for the improved performance of our model.

$$C_t = f_t \oplus i_t \quad (7)$$

Finally, the updated cell state C_t is sent to the $\tanh$ activation function, scaled to obtain a new candidate vector and multiplied by the gate signal z_o element by element to selectively obtain the final output $h_t \in R^{s \times n_1}$, which is then sent to the next unit.

$$h_t = z_o \odot \tanh(C_t) \quad (8)$$

As described by the aforementioned formulas, we send the sequence features Z into the LSTM, while recoding the

sequence features, and we can also determine the hidden states of features and fuse these hidden states in time series to obtain diverse features, thus further improving the representation of global features. Finally, the hidden states output by each unit of LSTM are combined, denoted as $\mathbf{y} = \{\mathbf{y}_t \mid t \in [1, T]\}$, and sent to the classification sub-model $F_c(\omega_c)$, achieving competitive accuracy on the three datasets.

With the development of FGVC, the computer vision community has developed rich fine-grained datasets covering diverse meta-categories, such as birds [3, 6], dogs [8], cars [12], etc. Table 1 lists the statistical information regarding these datasets. We can observe that the performance of most of them is nearly saturated, except for bird datasets as a meta-category. Specifically, the CUB-200-2011 dataset comprises a total of 200 bird categories, including 5994 training images and 5794 testing images. Each category contains approximately 30 training images. NABirds contains 555 bird species, 23,929 training images and 24633 test images. Notably, these two datasets were proposed some time ago, although they still do not achieve satisfactory accuracy, which shows that the fine-grained dataset with birds as meta-species is more challenging. Our benchmark weak supervised fine-grained dataset, Birds-267-2022, was developed in this context, which can promote future research in FGVC. Next, we introduce the Bird-267-2022 dataset in detail.

3.2 Data collection and preprocessing

Most of the data we collected derive from the real shooting of wild scenes and <https://www.birdnet.cn/>. At the initial stage, a total of 14643 images from 298 categories were collected, and after strict screening, mainly including the removal of duplicate images and images with uncertain categories, the Bird-267-2022 dataset, which contains 267 categories and 12,233 images, was obtained, with the ratio of training set to test set at about 1:1.

Removal of duplicate images. The goal of machine learning is to obtain a model with good generalization ability, that is, a model that not only performs well on the training data but also achieves remarkable results on the test data. Therefore, we rigorously eliminated duplicate images between the training and test data. Our deduplication strategy assumes that images with similar semantic information are more likely to be similar or identical. First, we used the ResNet-50 model pre-trained on ImageNet-1K to extract embedded feature vectors from both the training and test sets. Specifically, we selected the feature maps of the last block output and then performed a global averaging pool to convert them into 2048-d feature vectors. Subsequently, for each test image in each category, we used the cosine simi-

larity function to calculate the similarity distance S between the test image and each training image. For each category, we obtained the minimum distance between the training and test data, which is expressed as $S_{\min}$. We set a threshold $\lambda = 0.3$, scaled the distance to $(1 + \lambda) \times S_{\min}$ and removed the training images with distances less than $(1 + \lambda) \times S_{\min}$. After removing duplicate images, the total number of images dropped to 12233.

3.3 Dataset statistics

The images in the Birds-267-2022 dataset were recorded in different places and environments, incorporating greater scale, illuminations, poses and viewpoints of objects, as well as category imbalance. Specifically, the following main aspects were considered when creating the Birds-267-2022 dataset.

Realistic field scene: The goal of deep learning is to apply it to the real world. In terms of FGVC, the birds in the wild environment are more complicated than the birds around us, owing to the inherent limitations of the wild environment. The Birds-267-2022 dataset contains bird photos captured by our outdoor infrared camera. Therefore, this dataset consists of the following characteristics. **Small ROI:** Some birds are small or far from the camera and can be difficult to identify even for humans. **Occlusion:** Birds can be occluded by a branch or the edge of a frame. **Motion blur:** The shutter speed of the camera is not sufficiently fast to eliminate motion blur; therefore, the images of birds are occasionally blurry. **Complex background:** Owing to various environments in the wild, the colors of some birds are extremely similar to those of their environments, rendering it difficult for humans to distinguish them. Considering the aforementioned aspects as well as the challenges in existing FGVC datasets, we proposed the Bird-267-2022 dataset for FGVC.

4 Bird-267-2022 benchmark

4.1 Advantages of the Bird-267-2022 dataset

In this section, we compare Birds-267-2022 dataset with the most commonly used fine-grained dataset, CUB-200-2011. Figure 6 shows the overall information regarding the Birds-267-2022 and CUB-200-2011 datasets. First, the Bird-267-2022 dataset is 445 more than CUB-200-2011 in total and 67 more categories in categories. We also compared the numbers in each category. We can observe that the sample distribution of CUB-200-2011 is relatively balanced, whereas the sample distribution of the Birds-267-2022 dataset is extremely uneven, which is more in line with the species distribution in nature. Finally, it also contains bird

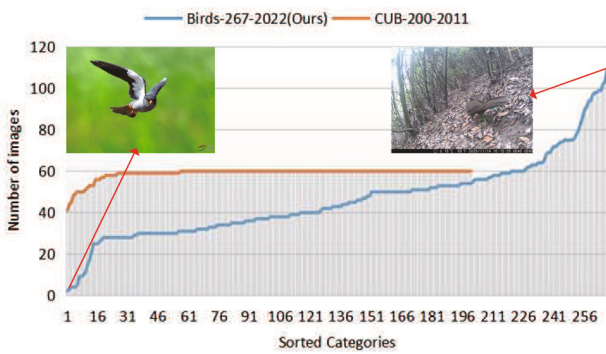


Fig. 6 Comparison between the Bird-267-2022 dataset and CUB-200-2011 dataset in terms of the number of categories obtained

pictures captured by our outdoor infrared camera, which is characterized by small-target objects and simulates a real scene. Therefore, the challenge of the dataset is increased compared with that of the CUB-200-2011 dataset.

5 Experiments

In this section, we systematically verify the effectiveness of the proposed method using two public FGVC datasets, i.e., CUB-200-2011 and NABirds, as well as the Birds-267-2022 dataset. We did not use any bounding box/part annotations. In Sect. 5.1, we introduce the settings of the experimental parameters. Furthermore, we introduce the quantitative results of the proposed method and compare them with those of the SotA methods. Finally, meaningful ablation experiments are introduced.

5.1 Experiments settings

In all of our experiments, we adjusted the training images to 384×384 and did not use any additional data enhancement technology during the training stage. In the testing stage, we scaled the size of the test images to 438×438 and then cropped the image to a final size of 384×384 using CenterCrop. We placed the LSTM at the end of the Swin Transformer and set the hidden state to 1536. The model was optimized by stochastic gradient descent (SGD) with a momentum of 0.9, decay of 0.0003, epoch number of 200, weight decay of 0.0005 and mini-batch of 6 using two Tesla T4 GPUs. The learning rate (LR) was adjusted using a cosine annealing scheduler. PyTorch was used to implement our algorithm. For all experiments, we used Top-1 accuracy as the evaluation standard. To verify the effectiveness of the proposed method, we compared some advanced methods, as listed in Table 2.

5.2 Quantitative results and comparison to the SotA methods

We compared the proposed method, TransFGVC, with the methods proposed by recent studies using the Bird-267-2022 and public datasets, CUB-200-2011 and NABirds. The experimental results for the Birds-267-2022, CUB-200-2011 and NABirds datasets are displayed in Tables 3, 4 and 6, respectively. From the results, we found that the performance of TransFGVC on these three datasets was better than most of all previous methods. In addition, we compare the methods involving LSTMs and make a simple analysis.

5.2.1 Performance analysis using the Birds-267-2022 dataset

We selected representative methods that are currently mainstream in FGVC and conducted experiments on the Birds-267-2022 dataset. For a fair comparison, all parameters are set objectively from the optimal values in the paper. The results are presented in Table 3. It can be concluded as follows.

- On the Birds-267-2022 dataset, the accuracy of the most advanced CAP method based on CNN reached 88.2%, whereas the accuracy of the TransFG and FFVT based on ViT reached 89.5% and 90.7%. Furthermore, the accuracy of PIM, HERBS and our TransFGVC based on Swin Transformer reached 91.9%, 91.7%, 91.5%, respectively, ranking in the top three. The preliminary results from Birds-267-2022 dataset show that the method based on ViT and its variants is more competitive than the method based on CNN in handling fine-grained problems. This may be because it can fully mine the distinguishing features by calculating the self-attention between fine-grained visual tokens to better deal with the small target problem in Birds-267-2022 dataset.
- Notably, at present, the highest accuracy of the dataset CUB-200-2011 is 93.1% of that of HERBS (Table 4), and that of the dataset Birds-267-2022 is 91.8% of that of PIM, which is 1.3% lower than that of CUB-200-2011, which shows that our dataset is more challenging than CUB-200-2011 dataset.

5.2.2 Performance analysis using the CUB-200-2011 dataset

We compared the proposed method, TransFGVC, with SotA using the CUB-200-2011 dataset. The CUB-200-2011 is a challenging and typical dataset for FGVC because of its high variance within the class and low variance between classes. The experimental results are listed in Table 4. From the table, we can draw the following conclusions.

Table 2 Partial baseline methods and their main techniques with publishing information

Baseline	Technique	Book title	Year
HERBS [7]	High-temperature refinement module and background suppression module, for extracting discriminative features and suppressing background noise	arxiv	2023
PIM [24]	A plugin module to output pixel-level feature maps and fuse filtered features	arxiv	2022
MetaFormer [23]	The joint learning of vision and various meta-information	arxiv	2022
TransFG [21]	ViT-based part selection module and a contrastive loss	AAAI	2022
TPSKG [45]	ViT architecture with peak suppression and knowledge guidance module	Neurocomputing	2022
HGNet [46]	Exploit the interconnection among hierarchical categories by hierarchical gate network	Neurocomputing	2022
CAP [11]	Context-aware attentional pooling module	AAAI	2021
FFVT [20]	ViT-based feature fusion and MAWS module	BMVC	2021
AP-CNN [30]	Attention pyramid convolutional network with dual-pathway hierarchy structure	TIP	2021
RAMS-Trans [47]	Recurrent attention multiscale ViT architecture combined with CNN	ACM	2021
FBSD [16]	Feature boosting, suppression, and diversification	IJCNN	2021
CAL [48]	A counterfactual attention learning method to learn more effective attention	ICCV	2021
PCA-Net [49]	A progressive co-attention network to recognize contrastive clues from different images	VCIP	2021
API-Net [19]	An attentive pairwise interaction network to identify differences by comparing image pairs	AAAI	2020

Table 3 Comparison of different methods using Birds-267-2022 dataset

Methods	Input resolution	Backbone	Pre-trained model	Accuracy(%)
SEF (SPL [50])	448 × 448	ResNet-50	ImageNet-1k	80.0
AP-CNN (TIP [30])	448 × 448	ResNet-50	ImageNet-1k	80.4
PMG (ECCV [51])	448 × 448	ResNet-50	ImageNet-1k	83.9
PCA-Net (VCIP [49])	448 × 448	ResNet-101	ImageNet-1k	84.7
MOMN (TIP [52])	448 × 448	Densenet-201	ImageNet-1k	82.2
FBSD (IJCNN [16])	448 × 448	Densenet-201	ImageNet-1k	85.2
MetaFormer (arxiv [23])	384 × 384	Metaformer-2	ImageNet-1k	85.0
CAP (AAAI [11])	224 × 224	Xception	ImageNet-1k	88.2
TransFGVC (Ours)	384 × 384	Swin-L	ImageNet-1k	90.1
MetaFormer (arxiv [23])	384 × 384	Metaformer-2	ImageNet-22k	89.6
TransFG (AAAI [21])	448 × 448	ViT-B_16	ImageNet-22k	89.5
FFVT (BMVC [20])	448 × 448	ViT-B_16	ImageNet-22k	90.7
TransFGVC (Ours)	384 × 384	Swin-L	ImageNet-22k	91.5
HERBS (arxiv [7])	384 × 384	Swin-L	ImageNet-22k	<i>91.7</i>
PIM (arxiv [24])	384 × 384	Swin-L	ImageNet-22k	91.9

The top three methods based on Image-Net 1k and Image-Net 22k are highlighted in bolditalic, italic and bold, respectively

Table 4 Comparison of different methods using the CUB-200-2011 dataset

Methods	Input resolution	Backbone	model	Accuracy (%)
MC-loss (TIP [53])	448 × 448	ResNet-50	ImageNet-1k	87.3
ACNet (CVPR [54])	448 × 448	ResNet-50	ImageNet-1k	88.1
AP-CNN (TIP [30])	448 × 448	ResNet-50	ImageNet-1k	88.4
HGNet (Neurocomputing [46])	448 × 448	ResNet-50	ImageNet-1k	88.7
PMG (ECCV [51])	448 × 448	ResNet-50	ImageNet-1k	89.6
Mix+ (VCIP [22])	448 × 448	ResNet-50	ImageNet-1k	90.2
FDL (AAAI [31])	448 × 448	DenseNet-161	ImageNet-1k	89.1
FBSD (IJCNN [16])	448 × 448	DenseNet-161	ImageNet-1k	89.5
API-Net (AAAI [19])	448 × 448	DenseNet-161	ImageNet-1k	90.0
Metaformer (arxiv [23])	384 × 384	Metaformer-2	ImageNet-1k	89.7
CIN (AAAI [55])	448 × 448	ResNet-101	ImageNet-1k	88.1
DBTNet (NeurIPS [56])	448 × 448	ResNet-101	ImageNet-1k	88.1
PCA-Net (VCIP [49])	448 × 448	ResNet-101	ImageNet-1k	88.9
SnapMix (AAAI [18])	448 × 448	ResNet-101	ImageNet-1k	89.6
CAL (ICCV [48])	448 × 448	ResNet-101	ImageNet-1k	90.6
CAP (AAAI [11])	224 × 224	Xception	ImageNet-1k	<i>91.8</i>
TransFGVC (Ours)	384 × 384	Swin-L	ImageNet-1k	92.3
RAMS-Trans (ACM [47])	448 × 448	ViT-B ₁₆	ImageNet-22k	91.3
TPSKG (Neurocomputing [45])	448 × 448	ViT-B ₁₆	ImageNet-22k	91.3
AFTrans (arxiv [57])	448 × 448	ViT-B ₁₆	ImageNet-22k	91.5
FFVT (BMVC [20])	448 × 448	ViT-B ₁₆	ImageNet-22k	91.6
TransFG (AAAI [21])	448 × 448	ViT-B ₁₆	ImageNet-22k	91.7
IELT (TMM [5])	448 × 448	ViT-B ₁₆	ImageNet-22k	91.8
SIM-Trans (ACMMM [58])	448 × 448	ViT-B ₁₆	ImageNet-22k	91.8
ViT-NeT (ICML [59])	224 × 224	Swin-B	ImageNet-22k	91.6
Metaformer (arxiv [23])	384 × 384	Metaformer-2	ImageNet-22k	91.8
DCAL (CVPR [60])	448 × 448	R50-ViT-Base	ImageNet-22k	92.0
TransFGVC (Ours)	384 × 384	Swin-L	ImageNet-22k	92.7
PIM (arxiv [24])	384 × 384	Swin-L	ImageNet-22k	92.8
HERBS (arxiv [7])	384 × 384	Swin-L	ImageNet-22k	93.1

The top three methods based on Image-Net 1k and Image-Net 22k are highlighted in bolditalic, italic and bold, respectively

- Overall, the proposed method surpassed all available advanced methods in the dataset of CUB-200-2011, except PIM and HERBS, which are based on a pre-trained model on ImageNet-22 K. The reason may be that PIM and HERBS add FPN (feature pyramid network) and PAN (path aggregation network), respectively, to fuse features from different scales of backbone networks. Advanced results have been achieved, but this undoubtedly increases the complexity of the network. For example, TransFGVC consists of 216.1M parameters and 103.0G FLOPs (Table 5). This is 15% params and 10% FLOPs (Table 5) less than the PIM; however, the accuracy is only 0.1% lower. Also, TransFGVC is lighter than HERBS, with 25% and 27% (Table 5) fewer params and FLOPs, and accuracy 0.4% lower than HERBS. The above analysis shows that TransFGVC is superior to all

methods except PIM and HERBS in accuracy. Although it is slightly lower than HERBS and PIM in accuracy, it is superior to them in calculation efficiency and scalability. In addition, our method has the effect of fine-grained feature enhancement and fusion compared with HERBS and PIM as shown in Fig. 8, which is our advantage over HERBS and PIM.

- By comparing different types of methods, once again, we found that the recently proposed FGVC methods based on ViT are extremely competitive compared to the previous method based on CNN. Further, the latest methods such as TransFGVC, PIM, HERBS based on the Swin Transformer outperform the existing methods based on ViT, which proves again that the transformer and its variants have clear advantages in solving FGVC problems.

Table 5 Calculation complexity and accuracy of TransFGVC in the Birds-267-2022, CUB-200-2011 and NABirds dataset are compared with the SotA approaches

Methods	Birds-267-2022 (%)	CUB-200-2011 (%)	NABirds (%)	Params (M)	FLOPs(G)
SIM-Trans (ACM MM [58])	–	91.80	–	88.2	118.1
IELT (TMM [5])	–	91.80	90.70	93.5	65.7
ViT-NeT (CML [59])	–	91.60	–	92.2	60.9
FFVT (BMVC [20])	90.70	91.60	–	86.4	256
TransFG (AAAI [21])	90.10	91.70	90.80	86.9	108.4
MetaFormer (arxiv [23])	89.60	91.80	92.20	80.6	50
HERBS (arxiv [7])	<i>91.70</i>	93.10	93.00	287.7	140.6
PIM (arxiv [24])	91.90	92.80	92.80	256.9	115.6
TransFGVC (ours)	91.50	92.70	91.40	216.1	103

Model complexity is measured by parameters (millions) and GFLOPs(billions). The top three methods based on Image-Net 22k are highlighted in bolditalic, italic and bold, respectively

Table 6 Comparison of different methods using the NABirds dataset

Method	Input resolution	Backbone	Pre-trained model	Accuracy (%)
API-Net (AAAI [19])	512×512	DenseNet-161	ImageNet-1k	88.1
HGNet (Neurocomputing [46])	448×448	ResNet-50	ImageNet-1k	86.4
GaRD (CVPR [61])	448×448	ResNet-50	ImageNet-1k	88.0
FixSENet-154 (NeurIPS [62])	448×448	SENet-154	ImageNet-1k	89.2
CAP (AAAI [11])	224×224	Xception	ImageNet-1k	<i>91.0</i>
TransFGVC (Ours)	384×384	Swin-L	ImageNet-1k	91.2
TPSKG (Neurocomputing [45])	448×448	ViT-B_16	ImageNet-22k	90.1
TransFG (AAAI [21])	448×448	ViT-B_16	ImageNet-22k	90.8
IELT (TMM [5])	448×448	ViT-B_16	ImageNet-22k	90.8
TransFGVC (Ours)	384×384	Swin-L	ImageNet-22k	91.4
Metaformer (arxiv [23])	384×384	Metaformer-2	ImageNet-22k	92.2
PIM (arxiv [24])	384×384	Swin-L	ImageNet-22k	92.8
HERBS (arxiv [7])	384×384	Swin-L	ImageNet-22k	93.0

The top three methods based on Image-Net 1k and Image-Net 22k are highlighted in bolditalic, italic and bold, respectively

- However, the Swin Transformer is extremely competitive in solving FGVC problems; the accuracy is improved by 1.12% after combining it with LSTM. This proves that LSTM can perform global modeling in FGVC.
- Compared with the methods based on a pre-trained model on Imagenet-1k, TransFGVC has surpassed the current SotA CNN-based method CAP by 0.2%. Although the size of our input image is 384×384 higher than that of CAP 224×224 , our batch size of 6 is lower than that of CAP's 12. Finally, we surpassed CAP in accuracy.
- Compared with methods based on models pre-trained on Imagenet-22k, the accuracy of TransFGVC ranks fourth. The main reason why it is lower than the SotA methods HERBS and PIM is that these methods focus on extracting rich multi-scale features of objects, which we have not targeted. In addition, the accuracy of TransFGVC is 0.8% lower than that of the SotA method MetaFormer, which is perhaps mainly due to a limitation of computing resources resulting in a lack of sufficient learning on a larger dataset. At the same time, TransFGVC surpasses

5.2.3 Performance analysis using the NABirds dataset

To fully verify the effectiveness of TransFGVC, we conducted an experiment on a more challenging dataset than CUB-200-2011, NABirds, which contains 505 categories and 48,562 images of North American birds. Because of its huge scale, only a small number of methods have been tested, and the proposed method outperformed most of the existing methods. The results are displayed in Table 6. The following conclusions can be drawn.

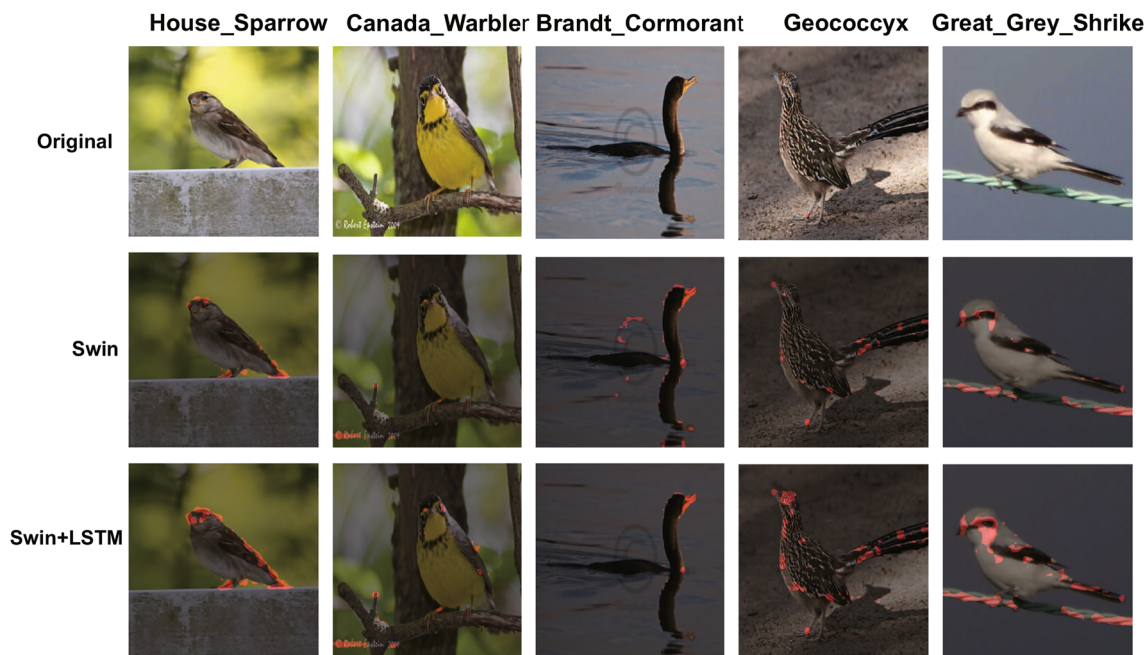


Fig. 7 Visualization results of the proposed method using the CUB-200-2011 dataset with the effect of fine-grained feature enhancement and fusion. The first row shows the original image, the second row shows the results of Swin-L, and the bottom row shows the results of Swin-L with LSTM

Metaformer by 1.9% and 0.9% on the Birds-267-2022 and CUB-200-2011 datasets, respectively.

5.2.4 Comparison with the algorithm involving LSTMs

Compared to CNN, LSTM is under-explored in FGVC. Therefore, we have found that three algorithms use LSTM to assist FGVC. We also performed a simple comparison with their methods. To the best of our knowledge, HSnet [63] was the first to use LSTM for FGVC, wherein a single-layer LSTM was used to capture long-term dependencies along the search trajectory. The accuracy reached 87.5% with the additional annotations. Although its accuracy is lower than that of the proposed method, we appreciate its valuable experience in FGVC. WS-CPM [9] is based on a CNN and uses a bidirectional LSTM to learn the rich representation of the features of all image patches. The accuracy of the proposed method for CUB-200-2011 was 2.2% higher than that of the WS-CPM. The CAP [11] uses a single-layer LSTM to capture the spatial arrangement of integral regions, which is 0.8% lower than the proposed method. We have created innovative explorations of the Swin Transformer combined with LSTM to implement global modeling among visual tokens. The aforementioned methods involving LSTM are all based on CNN, and we found that LSTM can also assist Swin Transformer and achieve advanced results. This may be because the segmented visual tokens contain more global information than the local area extracted by the CNN; therefore, LSTM

can achieve better global modeling and the goal of global information interaction.

5.3 Ablation study

To prove the validity of the proposed LSTM module, experimental analysis was performed for different variants of the module. The experiments were evaluated using CUB-200-2011 and our Birds-267-2022 dataset. We evaluated the influence of the following model designs on the experimental results: LSTM and its variants, parameter ablation, dimensions of LSTM hidden states and different pre-trained models.

5.3.1 Influence of LSTM and its variants

In this section, we explore the influence of a single-layer LSTM (L), a stacked LSTM (SL) and a bidirectional LSTM (BL) on baseline Swin Transformer(B) in the CUB-200-2011 and our Birds-267-2022 dataset. Details about them are found in appendix.

We make full use of the Swin Transformer to extract the discriminative part information. However, the experiment described in Sect. 5 shows that using only this information for FGVC is insufficient. According to our observations, existing ViT-based methods have two major problems. First, although ViT has shown significant potential for extracting discriminative features, considering these discriminative regions in

Table 7 Results of the proposed model with the addition of LSTM (L+B) and stacked LSTM (SL+B) or bidirectional LSTM (BL+B) to baseline Swin Transformer (B)

Evaluation	B	SL+B	BL+B	L+B
Top1	91.60%	92.30% (+ 0.7%)	92.40% (+ 1.0%)	92.72% (+ 1.12%)
The observed accuracy on CUB-200-2011 trend of top1 is (L+B) > (BL+B) > (SL+B) > B (the highest accuracy is shown in bold)				

**Fig. 8** Visualization results of the proposed method using the CUB-200-2011 data set have the effect of fine-grained feature enhancement and fusion compared with the SotA methods HERBS and PIM. The

first line shows the original image, the second line shows the results of TransFGVC (ours), and the last line shows the results of PIM

Table 8 Results of the proposed model with the addition of LSTM (L+B) and stacked LSTM (SL+B) or bidirectional LSTM (BL+B) to baseline Swin Transformer (B)

Evaluation	B	SL+B	BL+B	L+B
Top1	89.94%	90.76% (+ 0.82)	91.01% (+ 1.07%)	91.13% (+ 1.19%)
The observed accuracy on Birds-267-2022 trend of top1 is (L+B) > (BL+B) > (SL+B) > B (the highest accuracy is shown in bold)				

isolation renders it insufficient to handle fine-grained problems [21]. We believe that this is because of a lack of combination of contextual features (Fig. 4a). Second, in some cases, the extracted discriminative fine-grained features are still insufficient, particularly for complex backgrounds, and

they are easily overwhelmed by noise [20]. Therefore, we believe that the extracted fine-grained features require further unified enhancements (Fig. 4b). Therefore, we suggest an LSTM or variant model with a selective forgetting and memorizing function to receive and re-encode these sequence

Table 9 Effect of a single LSTM network on model performance under different hidden neuron dimensions using the CUB-200-2011 dataset

Hidden neuron dimensions	1024	1280	1536	1792	2048
Top1	92.58%	92.48%	92.72%	92.53%	92.39%

Table 10 Effect of a single LSTM network on model performance under different hidden neuron dimensions using the Birds-267-2022 dataset

Hidden neuron dimensions	1024	1280	1536	1792	2048
Top1	91.49%	91.37%	91.52%	91.50%	91.37%

tokens to further fuse and enhance the context features for FGVC.

The results in Tables 7 and 8 indicate that the three methods can improve the performance of Swin Transformer. This indicates that LSTM and its variants can promote FGVC owing to their global modeling ability. Moreover, the LSTM with a single layer exhibited the best performance. To further explain the necessity of combining Swin Transformer with LSTM, similar to Fig. 5, we provide more statistical significance test results and again compare the visual effect of our method with that of Swin Transformer. We used Grad-CAM to generate the heat maps. As shown in Fig. 7, from the categories Canada_Warbler and House_Sparrow in the first and second columns, respectively, it can be observed that the proposed method equally focuses on discriminative areas, such as the eye and forehead of the object. Compared to the baseline, it can effectively hold information on different parts. Fusion of the model avoids the common overfitting problem caused by the model's excessive focus on the most salient regions. From the categories Great_Grey_Shrike and Geococcyx in the fourth and fifth columns, respectively, it can be clearly observed that the proposed method enhances salient regions such as naps, eyes, tails and beaks. Simultaneously, it can further strengthen the fusion of contextual information. This can effectively alleviate the negative effects of noise in complex backgrounds. The conclusion drawn from the middle column is that, owing to the influence of the control signals sent by the LSTM forget gate and memory gate, our method can remove irrelevant information to a certain extent, preserving clean features for classification.

5.3.2 Influence of the dimension of LSTM hidden states

It is very important to choose an appropriate number of hidden neurons for LSTM. As shown in Tables 9 and 10, we found that it was difficult to achieve the expected effect with an excessively high or extremely low number of neurons. We believe that using extremely few neurons in the hidden layer will lead to the problem of underfitting. By contrast, when an excessively high number of neurons are used in the hidden layer, the limited information contained in the training data is insufficient to train all the neurons of hidden layer; therefore, an excessively high number of neurons in the hidden layer may lead to overfitting. Finally, based on the experimental results, we determined that the number of hidden neurons in the LSTM was 1536.

5.3.3 Influence of LR parameters

The influence of hyperparameters on the model cannot be ignored. In this study, we focused on adjusting the LR. As listed in Tables 11 and 12, we validate this and obtain a LR $3e-4$ and $2e-3$ that is appropriate for the proposed model

Table 11 Comparison of the proposed method under different LRs using the CUB-200-2011 dataset (the highest accuracy is shown in bold)

Lr	1e-4	2e-4	3e-4	4e-4	5e-4
Top1	92.42%	92.42%	92.72%	92.54%	92.46%

Table 12 Comparison of the proposed method under different LRs using the Birds-267-2022 dataset (the highest accuracy is shown in bold)

Lr	1e-3	2e-3	3e-3	4e-3	5e-3
Top1	91.27%	91.52%	91.26%	91.13%	90.65%

Table 13 Comparison of the proposed method under different pre-trained models using the CUB-200-2011 dataset (the highest accuracy is shown in bold)

Pre-trained model	ImageNet-1K	ImageNet-22K
Top1	92.30%	92.70%

Table 14 Comparison of the proposed method under different pre-trained models using the Birds-267-2022 dataset (the highest accuracy is shown in bold)

Pre-trained model	ImageNet-1K	ImageNet-22K
Top1	90.10%	91.50%

in the CUB-200-2011 and Birds-267-2022 dataset, respectively.

5.3.4 Influence of different pre-training models

As listed in Tables 13, 14 and 15, we compared the effect of different scales of pre-trained models migrating to the FGVC; we found that our method TransFGVC based on a pre-trained model of ImageNet-22K outperforms the pre-trained model based on ImageNet-1K by 1.4%, 0.4% and 0.2% on the Birds-267-2022, CUB-200-2011 and NABirds datasets, respectively.

Table 15 Comparison of the proposed method under different pre-trained models using the NABirds dataset (the highest accuracy is shown in bold)

Pre-trained model	ImageNet-1K	ImageNet-22K
Top1	91.20%	91.40%

6 Conclusion

In this study, we proposed a novel FGVC algorithm, TransFGVC, and developed a new class-unbalanced FGVC dataset, Birds-267-2022. The proposed algorithm achieved 92.7%, 91.4% and 91.5% accuracy using the CUB-200-2011, NABirds and Birds-267-2022 datasets, respectively. The efficient fusion of the Swin Transformer and LSTM enabled the TransFGVC algorithm to deliver competitive performance across all three datasets. The research results of this paper will help to promote the practical application of FGVC algorithms and provide insight to the application of LSTM in the field of computer vision. Nonetheless, transformer-based methods generally have the problem of requiring a large amount of computation. Reducing the amount of computation without reducing accuracy will be the key goal of our future research. In addition, small-target identification and class imbalance in FGVC remain open research directions.

Appendix: Supplementary Material

A. Failure samples analysis

Figure 9 shows the number of prediction error samples in each class in the data set of CUB-200-2011. A number of 0 indicates that all the samples in this class are predicted correctly. Among them, there are more prediction errors in several classes, such as 16, 15 and 12 samples, which

need to be paid attention to and solved in the future. In order to more clearly count the number of wrong predictions of each class into other classes, we made the confusion matrix as shown in Fig. 10. The numbers on the main diagonal of the confusion matrix represent the correct number of samples predicted by each class, and the non-diagonal numbers on each row represent the number of samples wrongly predicted by the current class into other classes, which is convenient for future comparative analysis and research to further improve the model performance.

B. Swin Transformer architecture

Patch embedding. Consistent with the application of the Swin Transformer, we uniformly adjust the original image resolution to $H \times W$ and then divide nonoverlapping patches using a sliding window with step size S to obtain $N = H \times W / S^2$ patches, denoted as $P = \{P_n | n \in [1, N]\}$. Each patch is then flattened and mapped to the C dimension through a learnable linear projection to generate visual tokens. The visual tokens are denoted as $X = \{X_n | n \in [1, N]\}$. Finally, it is sent to the transformer block.

Swin Transformer block. The Swin Transformer block consists of self-attention and stacked MLP layers. In the Swin Transformer, self-attention is divided into window-based multi-head self-attention (W-MSA) and shifted window based multi-head self-attention (SW-MSA). All attention is calculated within the divided windows. This

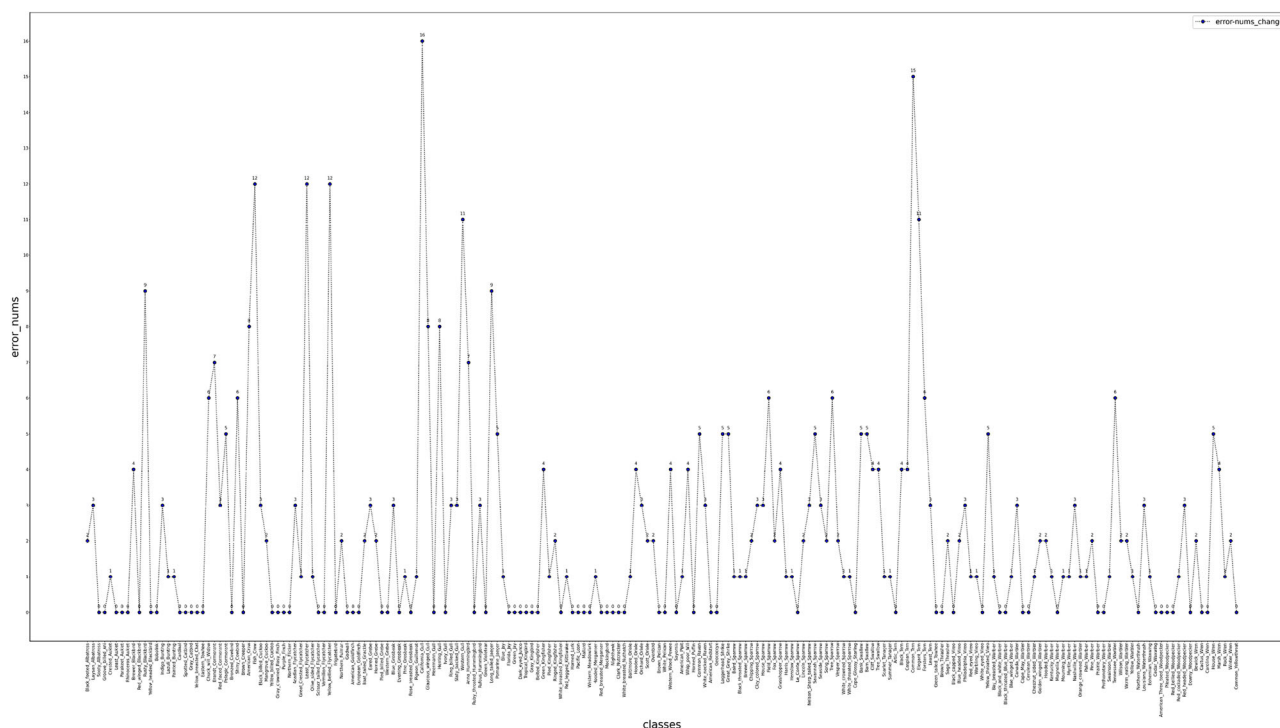


Fig. 9 Line chart of the number of prediction errors of each class on the data set of CUB-200-2011

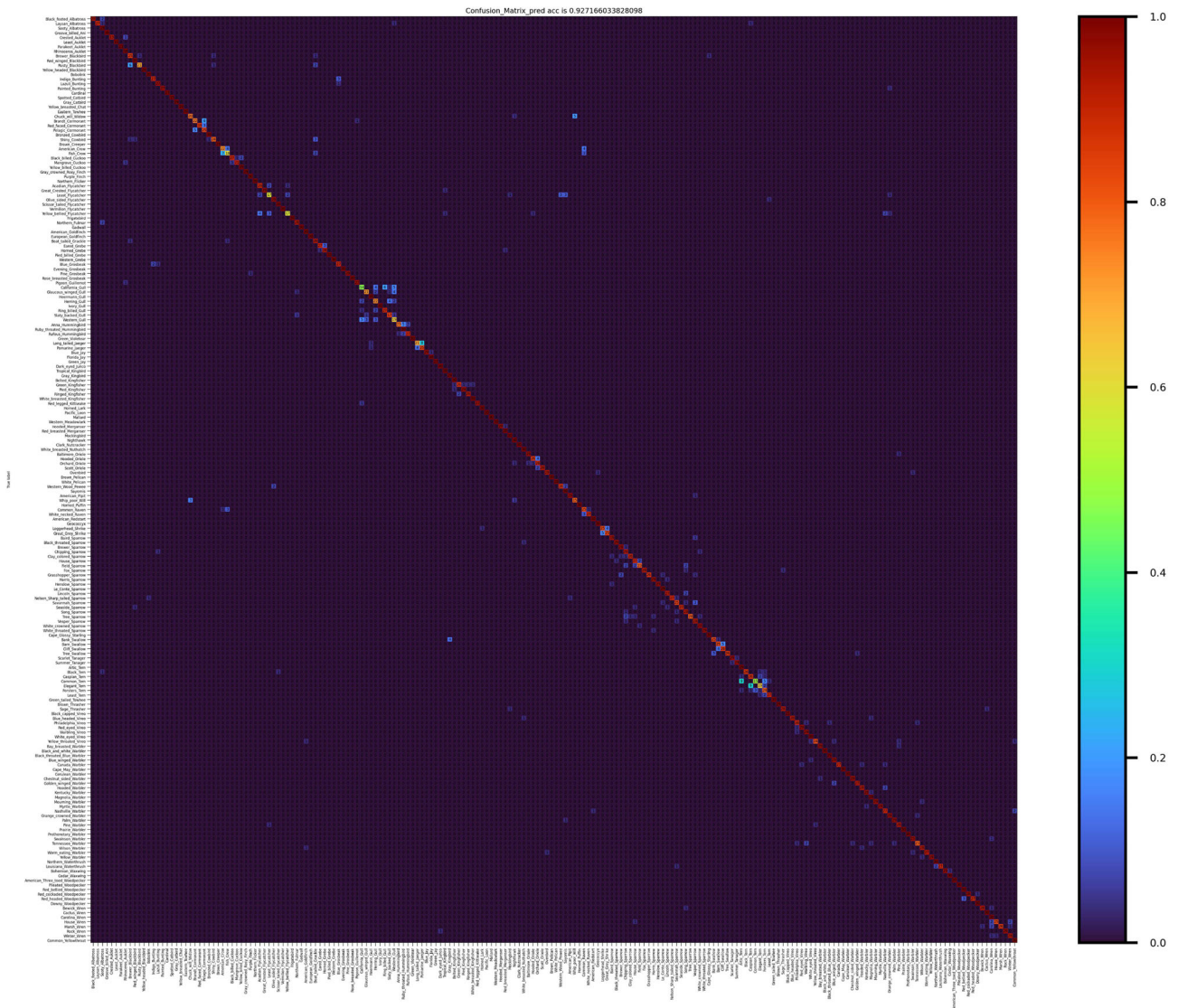


Fig. 10 Confusion matrix formed by prediction results of each class on CUB-200-2011 data set

significantly reduces the computational overhead. SW-MSA adds a shift operation based on W-MSA. The function is to further introduce cross-window connections while maintaining the efficient computation of nonoverlapping windows. In other words, it can also increase the interaction between different parts of an object. The SW-MSA and W-MSA appear alternately in consecutive blocks. If the window is properly divided, the continuous Swin Transformer block can be calculated as expressed by Eq. (1). We assume that total L blocks exist; l represents the l^{th} block. After the stacking of multiple blocks, we finally obtain the encoded T tokens, denoted as $\mathbf{Z} = \{\mathbf{Z}_t \mid t \in [1, T]\}$.

- C. LSTM layer *LSTM* for feature fusion and enhancement. Here, we attempt to use a single-layer LSTM module $F_l(\omega_l)$ for fine-grained feature enhancement and fusion.

The structure of the single-layer LSTM is shown in Fig. 11. First, the divided patches of P are fed into the Swin Transformer for training to obtain T tokens, which are marked as $\mathbf{Z} = \{\mathbf{Z}_t \mid t \in [1, T]\}$ in the previous section. These tokens are then considered to be sequences sent to each unit of the LSTM by running the LSTM to re-encode and fuse the tokens of each unit. Then, we concatenate the output of each unit, denoted as $\mathbf{y} = \{\mathbf{y}_t \mid t \in [1, T]\}$, to obtain fused and enhanced context features. Based on the experiment described in Sect. 5.3.1, we select a single-layer LSTM. Based on the experiment described in Sect. 5.3.2, the hidden state was set to 1536. The complete workflow of LSTM is described in detail in Sect. 3.1.

- D. LSTM and its variants

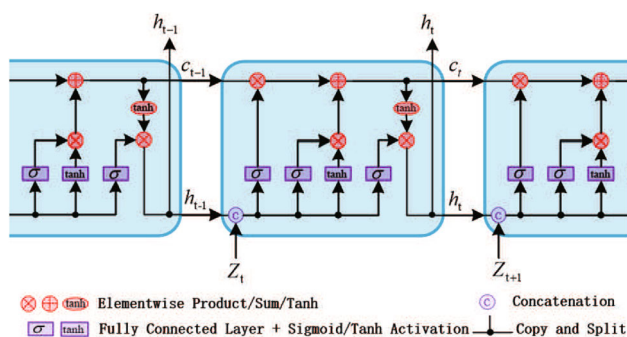


Fig. 11 Single LSTM Layer

- Single-layer LSTM: **L** was the first consideration for usage in this study. We set the input and hidden dimensions to 1536, which were the same as **SL** and **DL**.
- Stacked LSTM: **SL** is formed by stacking double-layer LSTMs, with the second LSTM accepting the outputs of the first LSTM and computing the results.
- Bidirectional LSTM: **BL** has two recurrent layers: the input of the forward layer receives the original input sequence, and the reverse layer receives the reverse copy of the original input sequence. Based on experience, we concat the output value of the corresponding positions of the two layers as the result.

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Data availability The data that support the findings of this study are available from the corresponding author, [LF Shen], upon reasonable request.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Khan, S.D., Ullah, H.: A survey of advances in vision-based vehicle re-identification. *Comput. Vis. Image Understand.* **182**, 50–63 (2019)
2. Karlinsky, L., Shtok, J., Tzur, Y., Tzadok, A.: Fine-grained recognition of thousands of object categories with single-example training. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4113–4122 (2017)
3. Van Horn, G., Branson, S., Farrell, R., Haber, S., Barry, J., Ipeirotis, P., Perona, P., Belongie, S.: Building a bird recognition app and large scale dataset with citizen scientists: the fine print in fine-grained dataset collection. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 595–604 (2015)
4. Nilsback, M.-E., Zisserman, A.: Automated flower classification over a large number of classes. In: *2008 Sixth Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)*, pp. 722–729 (2008) IEEE
5. Xu, Q., Wang, J., Jiang, B., Luo, B.: Fine-grained visual classification via internal ensemble learning transformer. *IEEE Trans. Multimed.* (2023). <https://doi.org/10.1109/TMM.2023.3244340>
6. Wah, C., Branson, S., Welinder, P., Perona, P., Belongie, S.: The Caltech-UCSD birds-200-2011 Dataset. California Institute of Technology (2011)
7. Chou, P.-Y., Kao, Y.-Y., Lin, C.-H.: Fine-grained Visual Classification with High-temperature Refinement and Background Suppression (2023)
8. Khosla, A., Jayadevaprakash, N., Yao, B., Li, F.-F.: Novel dataset for fine-grained image categorization: Stanford dogs. In: *Proceedings of CVPR Workshop on Fine-Grained Visual Categorization*, vol. 2(1). Citeseer (2011)
9. Ge, W., Lin, X., Yu, Y.: Weakly supervised complementary parts models for fine-grained image classification from the bottom up. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3034–3043 (2019)
10. Parkhi, O.M., Vedaldi, A., Zisserman, A., Jawahar, C.V.: Cats and dogs. In: *2012 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3498–3505 (2012). <https://doi.org/10.1109/CVPR.2012.6248092>
11. Behera, A., Wharton, Z., Hewage, P., Bera, A.: Context-aware attentional pooling (cap) for fine-grained visual classification. *AAAI* (2021)
12. Krause, J., Stark, M., Deng, J., Fei-Fei, L.: 3d object representations for fine-grained categorization. In: *Proceedings of the IEEE International Conference on Computer Vision Workshops (ICCV)*, pp. 554–561 (2013)
13. Liu, D., Zhao, L., Wang, Y., Kato, J.: Learn from each other to classify better: cross-layer mutual attention learning for fine-grained visual classification. *Pattern Recognit.* **140**, 109550 (2023)
14. Maji, S., Rahtu, E., Kannala, J., Blaschko, M., Vedaldi, A.: Fine-grained visual classification of aircraft (2013). arXiv preprint [arXiv:1306.5151](https://arxiv.org/abs/1306.5151)
15. Bossard, L., Guillaumin, M., Gool, L.V.: Food-101-mining discriminative components with random forests. In: *European Conference on Computer vision (ECCV)*, pp. 446–461. Springer (2014)
16. Song, J., Yang, R.: Feature boosting, suression, and diversification for fine-grained visual classification. In: *2021 International Joint Conference on Neural Networks (IJCNN)*, pp. 1–8 (2021). <https://doi.org/10.1109/IJCNN52387.2021.9534004>
17. Chou, P.-Y., Lin, C.-H., Kao, W.-C.: A novel plug-in module for fine-grained visual classification (2022). arXiv preprint [arXiv:2202.03822](https://arxiv.org/abs/2202.03822)
18. Huang, S., Wang, X., Tao, D.: Snapmix: semantically proportional mixing for augmenting fine-grained data, vol. 35(2), pp. 1628–1636 (2021)
19. Zhuang, P., Wang, Y., Qiao, Y.: Learning attentive pairwise interaction for fine-grained classification. In: *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 34, pp. 13130–13137 (2020)

20. Wang, J., Yu, X., Gao, Y.: Feature fusion vision transformer for fine-grained visual categorization (2021). arXiv preprint [arXiv:2107.02341](https://arxiv.org/abs/2107.02341)
21. He, J., Chen, J.-N., Liu, S., Kortylewski, A., Yang, C., Bai, Y., Wang, C.: Transfg: a transformer architecture for fine-grained recognition. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 36, pp. 852–860 (2022)
22. Li, H., Zhang, X., Tian, Q., Xiong, H.: Attribute mix: semantic data augmentation for fine grained recognition. In: 2020 IEEE International Conference on Visual Communications and Image Processing (VCIP), pp. 243–246. IEEE (2020)
23. Diao, Q., Jiang, Y., Wen, B., Sun, J., Yuan, Z.: MetaFormer: a unified meta framework for fine-grained recognition (2022)
24. Chou, P.-Y., Lin, C.-H., Kao, W.-C.: A novel plug-in module for fine-grained visual classification (2022)
25. Li, M., Lei, L., Sun, H., Li, X., Kuang, G.: Fine-grained visual classification via multilayer bilinear pooling with object localization. *Vis. Comput.* **8**, 1–10 (2022)
26. Guo, C., Lin, Y., Xu, M., Shao, M., Yao, J.: Inverse transformation sampling-based attentive cutout for fine-grained visual recognition. *Vis. Comput.* **8**, 1–12 (2022)
27. Zhang, N., Donahue, J., Girshick, R., Darrell, T.: Part-based R-CNNs for fine-grained category detection. In: European Conference on Computer Vision (ECCV), pp. 834–849. Springer (2014)
28. Chai, Y., Lempitsky, V., Zisserman, A.: Symbiotic segmentation and part localization for fine-grained categorization. In: Proceedings of the IEEE International Conference on Computer Vision (ICCV), pp. 321–328 (2013). <https://doi.org/10.1109/ICCV.2013.47>
29. Huang, S., Xu, Z., Tao, D., Zhang, Y.: Part-stacked CNN for fine-grained visual categorization. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 1173–1182 (2016). <https://doi.org/10.1109/CVPR.2016.132>
30. Ding, Y., Ma, Z., Wen, S., Xie, J., Chang, D., Si, Z., Wu, M., Ling, H.: AP-CNN: weakly supervised attention pyramid convolutional neural network for fine-grained visual classification. *IEEE Trans. Image Process.* **30**, 2826–2836 (2021). <https://doi.org/10.1109/TIP.2021.3055617>
31. Liu, C., Xie, H., Zha, Z.-J., Ma, L., Yu, L., Zhang, Y.: Filtration and distillation: enhancing region attention for fine-grained visual categorization. In: Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), vol. 34, pp. 11555–11562 (2020)
32. Zheng, H., Fu, J., Zha, Z.-J., Luo, J.: Looking for the devil in the details: learning trilinear attention sampling network for fine-grained image recognition. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2019)
33. Chen, W., Liu, T.-y., Lan, Y., Ma, Z.-m., Li, H.: Ranking measures and loss functions in learning to rank. In: Advances in Neural Information Processing Systems, vol. 22 (2009)
34. Kong, S., Fowlkes, C.: Low-rank bilinear pooling for fine-grained classification. In: 2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 7025–7034 (2017). <https://doi.org/10.1109/CVPR.2017.743>
35. Lin, T.-Y., RoyChowdhury, A., Maji, S.: Bilinear CNN models for fine-grained visual recognition. In: 2015 IEEE International Conference on Computer Vision (ICCV), pp. 1449–1457 (2015). <https://doi.org/10.1109/ICCV.2015.170>
36. Gao, Y., Beijbom, O., Zhang, N., Darrell, T.: Compact bilinear pooling. In: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 317–326 (2016). <https://doi.org/10.1109/CVPR.2016.41>
37. Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B.: Swin transformer: hierarchical vision transformer using shifted windows. In: International Conference on Computer Vision (ICCV) (2021)
38. Branson, S., Van Horn, G., Belongie, S., Perona, P.: Bird species categorization using pose normalized deep convolutional nets. In: BMVC, pp. 1–14 (2014)
39. Xie, L., Wang, J., Zhang, B., Tian, Q.: Incorporating visual adjectives for image classification. *Neurocomputing* **182**, 48–55 (2016). <https://doi.org/10.1016/j.neucom.2015.12.008>
40. Zheng, H., Fu, J., Mei, T., Luo, J.: Learning multi-attention convolutional neural network for fine-grained image recognition. In: Proceedings of the IEEE International Conference on Computer Vision (ICCV), pp. 5209–5217 (2017)
41. Girshick, R., Donahue, J., Darrell, T., Malik, J.: Rich feature hierarchies for accurate object detection and semantic segmentation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 580–587 (2014)
42. Cui, Y., Zhou, F., Wang, J., Liu, X., Lin, Y., Belongie, S.: Kernel pooling for convolutional neural networks. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 2921–2930 (2017)
43. Cai, S., Zuo, W., Zhang, L.: Higher-order integration of hierarchical convolutional activations for fine-grained visual categorization. In: Proceedings of the IEEE International Conference on Computer Vision (CVPR), pp. 511–520 (2017)
44. Engin, M., Wang, L., Zhou, L., Liu, X.: Deepkspd: learning kernel-matrix-based SPD representation for fine-grained image recognition. In: Proceedings of the European Conference on Computer Vision (ECCV), pp. 612–627 (2018)
45. Liu, X., Wang, L., Han, X.: Transformer with peak suppression and knowledge guidance for fine-grained image recognition. *Neurocomputing* **492**, 137–149 (2022)
46. Chen, Y., Song, J., Song, M.: Hierarchical gate network for fine-grained visual recognition. *Neurocomputing* **470**, 170–181 (2022)
47. Hu, Y., Jin, X., Zhang, Y., Hong, H., Zhang, J., He, Y., Xue, H.: Rams-trans: recurrent attention multi-scale transformer for fine-grained image recognition, pp. 4239–4248 (2021)
48. Rao, Y., Chen, G., Lu, J., Zhou, J.: Counterfactual attention learning for fine-grained visual categorization and re-identification, pp. 1025–1034 (2021)
49. Zhang, T., Chang, D., Ma, Z., Guo, J.: Progressive co-attention network for fine-grained visual classification, pp. 1–5. IEEE (2021)
50. Luo, W., Zhang, H., Li, J., Wei, X.-S.: Learning semantically enhanced feature for fine-grained image classification. *IEEE Signal Process. Lett.* **27**, 1545–1549 (2020)
51. Du, R., Chang, D., Bhunia, A.K., Xie, J., Ma, Z., Song, Y.-Z., Guo, J.: Fine-grained visual classification via progressive multi-granularity training of jigsaw patches. In: European Conference on Computer Vision (ECCV), pp. 153–168. Springer (2020)
52. Min, S., Yao, H., Xie, H., Zha, Z.-J., Zhang, Y.: Multi-objective matrix normalization for fine-grained visual recognition. *IEEE Trans. Image Process.* **29**, 4996–5009 (2020)
53. Chang, D., Ding, Y., Xie, J., Bhunia, A.K., Li, X., Ma, Z., Wu, M., Guo, J., Song, Y.-Z.: The devil is in the channels: mutual-channel loss for fine-grained image classification. *IEEE Trans. Image Process.* **29**, 4683–4695 (2020)
54. Ji, R., Wen, L., Zhang, L., Du, D., Wu, Y., Zhao, C., Liu, X., Huang, F.: Attention convolutional binary neural tree for fine-grained visual categorization. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 10468–10477 (2020)
55. Gao, Y., Han, X., Wang, X., Huang, W., Scott, M.: Channel interaction networks for fine-grained image categorization. In: Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), vol. 34, pp. 10818–10825 (2020)
56. Zheng, H., Fu, J., Zha, Z.-J., Luo, J.: Learning deep bilinear transformation for fine-grained image representation. In: Advances in Neural Information Processing Systems (NIPS), vol. 32 (2019)

57. Zhang, Y., Cao, J., Zhang, L., Liu, X., Wang, Z., Ling, F., Chen, W.: A free lunch from VIT: adaptive attention multi-scale fusion transformer for fine-grained visual recognition, pp. 3234–3238. IEEE (2022)
58. Sun, H., He, X., Peng, Y.: Sim-trans: structure information modeling transformer for fine-grained visual categorization. In: Proceedings of the 30th ACM International Conference on Multimedia (ACM), pp. 5853–5861 (2022)
59. Kim, S., Nam, J., Ko, B.C.: Vit-net: interpretable vision transformers with neural tree decoder. In: International Conference on Machine Learning (ICML), pp. 11162–11172. PMLR (2022)
60. Zhu, H., Ke, W., Li, D., Liu, J., Tian, L., Shan, Y.: Dual cross-attention learning for fine-grained visual categorization and object re-identification. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 4692–4702 (2022)
61. Zhao, Y., Yan, K., Huang, F., Li, J.: Graph-based high-order relation discovery for fine-grained recognition. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 15079–15088 (2021)
62. Touvron, H., Vedaldi, A., Douze, M., Jégou, H.: Fixing the train-test resolution discrepancy. In: Advances in Neural Information Processing Systems (NIPS), vol. 32 (2019)
63. Lam, M., Mahasseni, B., Todorovic, S.: Fine-grained recognition as hsnet search for informative image parts. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 2520–2529 (2017)

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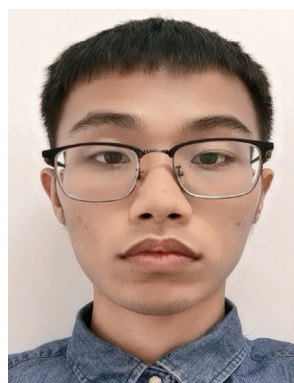
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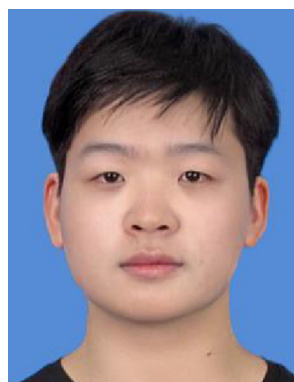
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